Group 1 Progress Report: Detecting Parkinson's Disease Using Acoustic Characteristics

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1 Introduction

Parkinson's disease (PD) is a progressive neurodegenerative disorder that affects motor function and speech. Clinical assessment involves identifying bradykinesia, or slowed movement, alongside rest tremor or rigidity (Postuma et al., 2015). One manifestation of this is hypokinetic dysarthria (HD), which involves reduced loudness, limited pitch variation, breathy or hoarse voice quality, and imprecise consonant production. These difficulties affect several aspects of speech production, including phonation, articulation, respiration, resonance, and prosody, providing a basis for investigating acoustic indicators of PD (Atalar et al., 2023). HD is reported to affect about 89% of people with PD (Byatt, 2025).

Our project investigates the following question: *Which acoustic characteristics are most useful for distinguishing Parkinson's disease from healthy controls, and which machine learning approach best captures these differences?* We aim to examine whether measurable properties of voice contain useful PD-associated patterns and to identify which characteristics contribute most to classification.

This work is motivated by the potential for low-cost, accessible assessment using voice recordings that require minimal equipment. PD is difficult to diagnose without a specialist or expensive tests, and it is reported to be misdiagnosed about 30% of the time (NCBI/NIH, 2020). Local healthcare access challenges reinforce this motivation: reported MRI wait times in Hamilton average about 170 days, and more remote places face longer waits alongside travel costs (Ontario Health, 2026). Although acoustic analysis cannot replace diagnostic imaging, it may offer a complementary approach to screening and monitoring. Our project explores this potential through the analysis of PD-associated voice characteristics.

2 Related Work

Previous research has investigated both engineered acoustic features and learned representations. Little et al. (2009) classified PD using dysphonia measurements from sustained phonations and used an SVM to distinguish participants with PD from healthy controls. Costantini et al. (2023) found comparable performance between feature-based classifiers and a convolutional neural network, whereas Iyer et al. (2023) achieved better performance with spectrogram-based Inception V3 transfer learning than with random forest and logistic regression models. Idrisoglu et al. (2023) identified SVMs as the most frequently used method across reviewed studies of voice-affecting disorders. Malekroodi et al. (2025) highlighted limitations involving small datasets and insufficient external or subject-independent validation in voice-based PD research. These findings motivate our focus on a compact acoustic feature set, a comparison between conventional classifiers, and participant-separated evaluation.

3 Dataset

We use the Mobile Device Voice Recordings dataset from King's College London (Trivedi et al., 2020), referred to as KCL from here on. It contains audio recordings of 37 participants (21 healthy controls, HC, and 16 people with PD) collected by Dhaval Trivedi, Hagen Jaeger and Michael Stadtschnitzer at King's College Hospital in London, UK. Each participant was recorded in a 10 m^2 room on a Motorola Moto G4 using a self-developed app similar to i-PROGNOSIS, between May 17, 2019 and January 24, 2020. Participants read two texts ("The North Wind and the Sun" and a technical snippet about computer applications in geography), which form the *Read Text* (RT) recordings. They also had a simple phone conversation, which forms the *Spontaneous Dia-*

	HC	PD	Total
Participants	21	16	37
Recordings (RT + SD)			73
Segments	595	418	1,013
Segments (% of total)	58.7	41.3	100

Table 1: Dataset after splitting into 10-second segments.

logue (SD) recordings; participant 18 has no SD recording. Recordings are separated into PD and HC folders, and each `.wav` file name encodes the participant ID, diagnosis, Hoehn and Yahr stage, and UPDRS II-5 and III-18 scores (for example, `ID32_pd_3_1_1`).

We did not use the numerical scores in the file names as features. They describe the progression of PD, so any non-zero value already implies that the participant has the disease, which would leak the label into the model inputs.

There are 73 recordings (37 RT and 36 SD) of roughly two minutes each, which is well below the 1,000 data points we considered a reasonable size for a classifier. We therefore split every recording into 10-second segments. Because a model will see the same voice in many segments, there is a risk of overfitting to individual speakers; we address this by keeping all segments of a participant together when splitting the data (Section 6.1).

Preprocessing. We first created a directory structure split by diagnosis, then by participant ID, with subfolders for the SD and RT segments. Each `.wav` file was loaded as a Pydub `AudioSegment`, which allows splitting by milliseconds, and cut into consecutive 10-second increments. The last piece of each recording is discarded if it is not longer than 8 seconds (an arbitrary threshold chosen as the minimum duration likely to contain enough information). Each participant is stored as a dictionary holding the diagnosis, the scores from the file name, and the paths to the RT and SD segment folders. Segments are numbered from 0 (`0.wav`, `1.wav`, ...). The result is 1,013 segments (Table 1), of which 595 are HC and 418 are PD, with 507 RT and 506 SD segments.

4 Features

After splitting the recordings, each segment is converted into numerical features using Praat, an open-source tool for speech analysis (Boersma et al., 2026). Praat can be run from the command line, so our Python code calls it on every segment (`.wav` file as input) and reads the features from its standard output, one per line. Extracting features for the whole dataset takes about an hour, so the output is stored in nested dictionaries (RT and SD, keyed by segment number), flattened with pandas `json_normalize`, and saved as a pickle and CSV file to avoid repeating the extraction. Because of this cost we selected a small set of features now, with the longer-term goal of engineering features from a larger pool of Praat measurements.

We use six engineered acoustic features describing voice periodicity, amplitude variation, and spectral characteristics:

- **CPPS**: smoothed cepstral peak prominence, the prominence of periodic structure in the voice signal.
- **HNR**: harmonics-to-noise ratio, harmonic energy relative to noise.
- **Local shimmer**: cycle-to-cycle variation in vocal amplitude (stored as a fraction).
- **Local shimmer in dB**: amplitude variation on a logarithmic scale.
- **LTAS slope and tilt**: long-term average spectrum measures of how acoustic energy is distributed across frequency. The slope compares the 0–1 kHz and 1–4 kHz bands; the tilt is the same contrast after fitting a trend line over 600–4,000 Hz.

These six features were chosen because they make up the Acoustic Voice Quality Index (AVQI), which clinicians use to assess dysphonia (Curtis, 2026). A score of 3.30 or greater from the following formula is generally taken to indicate a likely voice disorder (Curtis, 2026):

$$\begin{aligned}\mathrm{AVQI} = 2.8902\,\big(&4.152 - 0.177\,\mathrm{CPPS}\\ &- 0.006\,\mathrm{HNR} - 0.037\,\mathrm{Shim}_{\%}\\ &+ 0.941\,\mathrm{Shim}_{dB} + 0.01\,\mathrm{slope}\\ &+ 0.093\,\mathrm{tilt}\big)\end{aligned} \tag{1}$$

We hope the machine learning models can learn a relationship similar to this score.

Each segment is therefore a six-dimensional numerical vector paired with a binary label (PD or HC). We keep the participant ID so that segments can be grouped during evaluation. Since the representation is already compact, we initially retain all six features, and we plan further experiments on whether removing redundant features improves performance.

5 Implementation

We implemented the classification pipeline in Python, using pandas and NumPy for data preparation, scikit-learn for modelling, and Matplotlib for visualization. For this progress report we focused on an initial Support Vector Machine (SVM) and a majority-class baseline. Random Forest and k-Nearest Neighbours remain planned comparisons. We used a fixed random seed of 42 to make the cross-validation splits reproducible.

Data checks. We loaded the extracted measurements from `segment_features.csv` and checked that the expected feature, label, and metadata columns were present. Our preprocessing function checked for invalid identifiers, missing or non-numeric measurements, infinite values, duplicate records, and inconsistent diagnoses within participants. These checks identified no records requiring removal, so we retained all 1,013 segments from 37 participants. We also examined feature distributions and correlations. Although the two shimmer measurements are strongly correlated, we retained all six features for this initial experiment and did not remove extreme observations.

Inputs and labels. The input matrix contains CPPS, HNR, local shimmer, local shimmer in dB, LTAS slope, and LTAS tilt. Each segment has a binary target: 0 for healthy control and 1 for PD. Participant identifiers are stored separately for grouping during evaluation, while the recording task and segment identifiers are kept as metadata. These identifiers are excluded from the model inputs, and we checked that features, labels, and metadata remain aligned.

SVM. Our SVM uses a radial basis function (RBF) kernel, which lets it represent nonlinear relationships between acoustic measurements and class labels. We combine `StandardScaler` and scikit-learn's `SVC` in a single pipeline. For each fold, the scaler learns feature means and standard deviations only from the training data and applies the same transformation to the validation data, so information from the validation participants cannot influence standardization. We used fixed settings of $C = 1.0$, `gamma="scale"`, and an optimization tolerance of 0.001, with no class weighting, feature selection, or hyperparameter tuning. The parameter C controls the trade-off between a wide classification margin and penalizing margin violations, while gamma controls the influence of each training observation in the RBF kernel. The learning objective combines margin regularization with hinge loss, which penalizes incorrect predictions and predictions that fall within the margin.

Baseline and training procedure. For comparison we implemented a `DummyClassifier` that predicts the most frequent label in each training fold. Both approaches are evaluated with a shared function that creates a fresh estimator for each fold, fits it on the training segments, and records predictions for the held-out segments. All five SVM fits converged. We save fold assignments, evaluation metrics, predictions, model settings, package versions, and confusion matrices to support reproducibility. We have not yet implemented a feature importance analysis, which is planned.

6 Results and Evaluation

6.1 Evaluation strategy

We evaluate both models with five-fold stratified group cross-validation, keeping each participant's segments together and approximately preserving class proportions. We verified that training and validation participants never overlap, that both classes appear in every fold, and that every segment receives exactly one out-of-fold prediction. We do not yet have a separate test set, and no hyperparameters were tuned, so the results are preliminary. We report accuracy, balanced accuracy, and PD precision, recall and F1-score (PD is the positive class), as means and standard deviations across the five folds.

6.2 Cross-validation results

The majority-class baseline always predicts healthy control, achieving 58.7% $\pm$ 1.4% accuracy, 50.0% balanced accuracy, and zero PD recall (Table 2). The SVM achieves 64.2% $\pm$ 11.1% accuracy, 61.7% $\pm$ 8.8% balanced accuracy, 64.4% PD precision, 47.4% PD recall, and an F1-score of 0.530. This is a modest average improvement, with considerable variation across folds: fold accuracy ranges from 0.49 to 0.76.

The pooled confusion matrix shows 455 correctly classified healthy-control segments and 198 correctly classified PD segments, alongside 140 false positives and 220 false negatives. The missed PD segments highlight the need to improve sensitivity. These are segment-level results for unseen participants, with longer recordings contributing

Model	Accuracy	Bal. accuracy	PD precision	PD recall	PD F1
Majority baseline	0.587 ± 0.014	0.500 ± 0.000	0.000 ± 0.000	0.000 ± 0.000	0.000 ± 0.000
SVM (RBF)	0.642 ± 0.111	0.617 ± 0.088	0.644 ± 0.187	0.474 ± 0.046	0.530 ± 0.056

Table 2: Segment-level five-fold grouped cross-validation results (mean $\pm$ standard deviation across folds).

Task	Acc.	PD recall	HC spec.
Read text (RT)	0.655	0.556	0.718
Spontaneous (SD)	0.634	0.400	0.815

Table 3: SVM out-of-fold performance by recording task (segment level).

more to the pooled metrics. Given the 37 participants and the absence of independent or external testing, we consider the findings preliminary.

6.3 Exploratory and error analysis

We also analysed the extracted features and the SVM's out-of-fold predictions in a separate notebook, without changing the models. We first compared the six features between groups using participant-level means (Figure 1). CPPS was lower in PD (7.67 dB vs. 8.48 dB in HC) and the LTAS slope was steeper (-19.96 dB vs. -16.73 dB), the largest group difference among the features (about 0.7 standard deviations). Shimmer was slightly higher in PD, while HNR showed almost no difference. All distributions overlapped substantially, and we did not run significance tests, so these differences are suggestive only. The two shimmer features were nearly identical ($r = 0.96$) and both correlated strongly with HNR ($|r| \approx 0.9$), so the six features contain less independent information than their number suggests.

We then aggregated the SVM's segment predictions to the participant level by majority vote. Participant-level accuracy was 62.2% (23/37), with 7 of 16 PD and 16 of 21 HC participants classified correctly. This is only modestly above the 56.8% obtained by always predicting HC, so voting across segments did not improve on segment-level accuracy. Finally, we split the out-of-fold predictions by recording task (Table 3). Accuracy was similar across tasks, but PD recall was lower for spontaneous dialogue than for read text.

7 Feedback and Plans

Feedback received. At the proposal stage, the TAs we consulted (Yasmin and Zizo) told us that speech data is acceptable as long as we clearly define which aspect of speech we analyse, that there is no required number of data points as long as we explain what the data represents, and that combining several data sources is allowed. We therefore focus on voice-quality measures (CPPS, HNR, shimmer, and LTAS) from 10-second segments.

Plans. Our initial results show that the SVM beats the majority baseline only modestly and misses more than half of the PD segments. We plan to:

- implement Random Forest and k-Nearest Neighbours on the same participant-grouped folds, so that all models are compared under identical conditions;
- run a held-out permutation-importance analysis to answer which acoustic characteristics matter most, accounting for correlated features such as the two shimmer measures;
- test whether removing redundant features (for example, one of the shimmer features) helps;
- try limited hyperparameter tuning and class weighting to improve PD recall, using nested participant-grouped validation so that the outer folds stay untouched;
- report participant-level results alongside segment-level results, since the clinical question concerns a person rather than a 10-second clip, and compare models trained separately on read text and spontaneous dialogue;
- consider additional Praat features, and external data if access is granted, to study generalization beyond this single dataset.

Reflection. The large fold-to-fold variation (fold accuracy from 0.49 to 0.76) shows how strongly the results depend on which few participants are in the validation set, so we will report variability for every model and avoid over-interpreting small differences. Keeping each participant within one fold was essential, since random segment-level splits would let a model recognise voices instead of PD. The weak participant-level result suggests that the current six features carry limited PD signal on their own, which motivates the feature analysis and the larger feature pool.

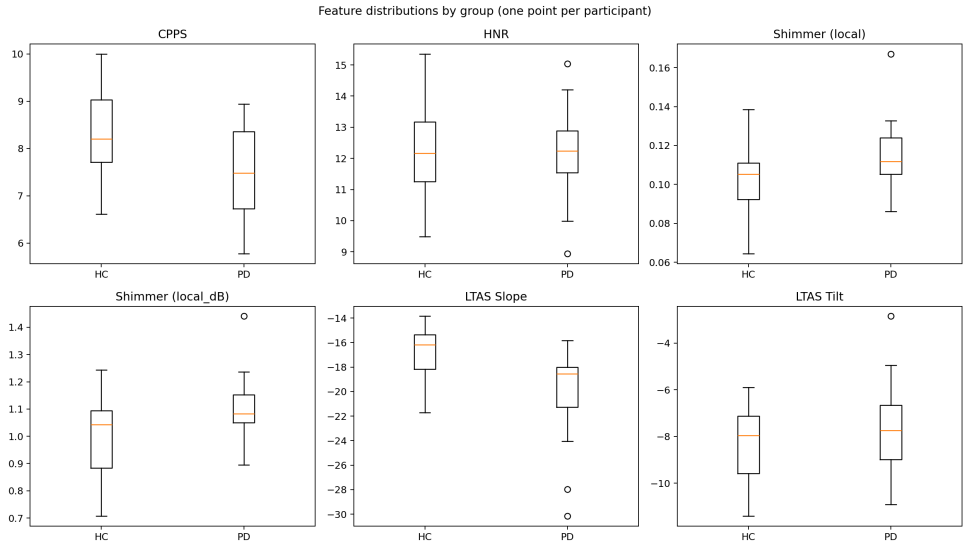


Figure 1: Distribution of each acoustic feature by group, with one point per participant (participant-level means).

Team Contributions

Nehan worked on the preprocessing and feature extraction for this project, including splitting the recordings and running Praat on the segments. He also wrote the introduction and the dataset and feature-extraction sections of this report.

Momina helped develop and implement the SVM classification pipeline, including data cleaning, participant-separated cross-validation, and the majority-class baseline. She also helped write and refine the progress report.

Yara carried out the exploratory analysis of the features and of the SVM predictions (PD versus HC feature comparison, participant-level and task-level results), wrote the corresponding results subsection, and merged the report sections in LaTeX.

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